Online Retail Analysis: Where Sales Come From, When They Rise, and Who Keeps Buying

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Executive Summary:

This report looks at a year of sales from a UK online gift shop to find out which are the largest export markets, which products are sold most by quantity and by revenue, when sales rise and which customers matter most. The data: UCI Online Retail dataset, covers about 540,000 transaction lines from 1 December 2010 to 9 December 2011.

Before any analysis, the data was cleaned. Duplicate rows were removed, missing product names were restored, and cancelled orders were set against the sales they cancelled. Every figure below is therefore net of returns. After cleaning, the data holds £9.76M of net revenue, 19,773 orders and 4,334 customers who could be identified. About a quarter of the rows had no customer ID, so the customer findings cover identified customers only.

What the analysis found

- The UK is the main market: It brings in 84.7% of net revenue (£8.27M). The five largest export markets are the Netherlands, Ireland, Germany, France and Australia, which together make up about 71% of the remaining £1.49M.
- Cancelled orders can mislead: Two products that looked like top sellers were mostly cancelled orders. One fell from £168,470 to £0 and the other from £81,701 to £4,221 once returns were counted. Among the 50 best-selling products, price explains very little of how much each sells: cheap items fill the top of the list, but being cheap does not make a best seller.
- Sales peak in autumn because more customers order : Revenue climbs from September and peaks in November at £1.42M, about double a typical summer month. Orders were 84% higher in November than the summer average, while the average order stayed about the same size. The timing fits Christmas stocking, and Easter shows no effect. However no particular pattern can be found since the data covers only one year and one week in total.
- Returning customers carry the business: Of the 4,334 identified customers, 2,829 (65%) ordered at least twice. They bring in 93.5% of the revenue from identified customers, about £2,730 each against about £360 for a one-time customer. Many of them might probably be shops or resellers restocking.

What the business should do: Protect returning customers before chasing new ones. Put export effort into the five markets named above. Judge products on net sales, and find out why two very large orders were cancelled.

What to keep in mind: The results rest on one year of data, so the link to Christmas rests on timing and is not proof. December 2011 covers only nine days. Many repeat buyers are probably wholesalers.

Objective and Scope:

The objective is to find where the business's revenue comes from, when it rises, and which customers matter most. The data is theUCI Online Retail dataset: 541,909 rows of transactions from a UK-based online gift retailer, from 1 December 2010 to 9 December 2011. Many of its customers might be wholesalers. Each row is one product line on one invoice, with the invoice number, stock code, description, quantity, date, unit price, customer ID and country. Source: Chen, D. (2015), UCI Machine Learning Repository, licensed CC BY 4.0.

Each candidate question was tested against four checks: can the columns answer it, is it specific, would the answer lead a business to act, and could the answer be surprising. Four questions remained.

1. How much revenue does each country bring in, with and without the UK?
2. Which products sell most, by quantity and by revenue, and do cheap products sell more?
3. When do sales rise, and does it match a festival?
4. How many customers are repeat buyers, and what do they buy?

Data and Method:

The data was cleaned before analysis. The raw data was kept untouched and all work was done on copies. Each decision below followed an inspection of the data.

Tools: Google Sheets (pivot tables, lookups, charts).

Definitions: net revenue is quantity × unit price, summed over sales and returns together. An order is a unique invoice number with a positive quantity. A repeat customer has two or more orders.

What was found	**What was done**	**Why**
5,268 Duplicate rows	Removed them	The same sale would be counted twice
1,454 rows with no product description	Filled them from other rows with the same stock code	Same code means same products, so the names can be recovered
About 135,000 rows with no customer ID	Kept them for country, product and monthly	The money was still earned, but the buyer cannot be

	questions: left time out of the customer question	identified
Negative Quantities	Set aside as cancellations, then added back as negative sales	Every total is then net of return
2,512 rows with a unit price of 0	Removed them	Free gifts or stock adjustments not sales
Postage, fees, manual entries and gift vouchers (codes such as POST, DOT, M, C2, BANK CHARGES, gift_)	Removed them	They are not products
One stock code with two description (popcorn holder)	Combined under one name	Otherwise its total is split into two lines

Two points came out of this step. Filling in the blank descriptions made the zero-price adjustment rows look like normal products, so they were re-examined before removal. And the popcorn holder appeared twice in a pivot, which showed that one product can carry two descriptions.

After cleaning, the data holds £9.76M of net revenue, 19,773 orders and 4,334 identified customers.

Findings:

Question 1: Where does the revenue come from?

The UK brings in 84.7% of net revenue (£8.27M of £9.76M). All other countries together bring in £1.49M.

The data has no revenue column, so one was created by multiplying quantity by unit price on every row. Because returns sit in the same table as negative sales, one pivot by country gives net revenue directly.

The UK is shown separately, because it is so large that it makes every other country look tiny on a chart. Among the export markets, five countries make up about 71% of the non-UK revenue.

Country	Net Revenue (£)	Share of Non-UK Revenue
Netherlands	283,480	19.0%
EIRE (Ireland)	259,380	17.4%

Germany	200,620	13.5%
France	182,077	12.2%
Australia	136,923	9.2%

Two entries, 'Unspecified' and 'European Community', are not single countries. They are kept in the totals but not counted as markets.

Implication: the Netherlands and Ireland are the biggest export markets, each ahead of Germany and France. Growth abroad is best started in these five markets. The business also depends heavily on one home market that is UK

Question 2: Which products sell most, by quantity and by revenue and do cheap ones sell more?

Two products that looked like the biggest sellers were mostly cancelled orders. This finding changed how the rest of the analysis was done.

A first pivot using sales only put 'Paper Craft, Little Birdie' (£168,470) and a ceramic storage jar (£81,701) near the top. With the returns added back, the first fell to £0 and the second to £4,221. A huge order had been placed and then cancelled. Without returns, the picture of what sells would have been wrong. From then on, every figure is net of returns.

Quantity and revenue give different lists.

	Product	**Net Total**
Most Units (quantity)	Popcorn holder (both sizes)	56,427 Units
	World war 2 gliders	53,751 Units
	Jumbo bag, red retrospot	46,256 Units
Most Revenue	Regency cakestand, 3 tier	£164,459
	White hanging heart T-shirt holder	£99,791
	Party bunting	£98,244

The gliders sell for about £0.25 each. The cakestand sells for about £12.65.

Do cheap products sell more? A little, but price explains very little. Among the 50 best sellers by quantity, most cost under £2.50, and the highest-volume items cost under £3. Yet many cheap products sell only modest volumes even inside that top 50, so being cheap does not make a product a top seller. Cheap items fill the top of the list, but price alone does not decide which products get there. Pricier items like the cakestand and the bunting earn their place through revenue, not volume.

The top 10 is used for the lists above. The quantities fall gradually after the first few products with no clear break, so 10 is a practical cut-off for reading, not a natural line.

Implication: keep cheap, high-volume products to keep orders flowing, keep a few higher-priced items that earn more per sale, and judge every product on net figures.

Question 3: When do sales rise?

Sales climb from September and peak in November at £1.42M, about double a typical summer month. The rise comes from more orders, not bigger ones.

Invoice dates were grouped by month in a pivot and net revenue was added up. From December 2010 to August 2011 revenue stayed between about £480k and £760k. Then it climbed to £1.01M in September, £1.06M in October and £1.42M in November.

The first hypothesis was that wholesalers were placing bigger orders to stock up. To test it, unique orders per month were counted, leaving out cancellations, and revenue was divided by orders. November had 2,751 orders against a May to August average of about 1,500, which is 84% more. Revenue per order only moved between £390 and £556 all year. So more customers ordered; they did not order more each.

Does it match a festival? The autumn build-up fits Christmas stocking. Easter fell on 24 April 2011, and April was the weakest month of the year, so Easter shows no bump. With one year of data the timing matches, but the data cannot show that Christmas is the cause.

The drop in December 2011 is not a real fall. The data stops on 9 December, and 816 orders in nine days is about 90 a day, close to November's pace.

Implication: stock and staff for more orders from September

Question 4: Who keeps buying?

Of the 4,334 customers identified, 2,829 (65%) ordered at least twice. They brought in 93.5% of the net revenue from identified customers.

Each customer's unique invoices were counted, not rows, because one order spans many rows. Cancellations were left out of this count: a customer with one order and one cancellation has not come back. A repeat buyer is defined as a customer with two or more invoices.

	Customers	Net revenue	Per customer
One-time	1,505 (34.7%)	£536,275 (6.5%)	About £356
Repeat, 2+ orders	2,829 (65.3%)	£7,725,183 (93.5%)	About £2,731

What do repeat customers buy? Mostly the products that lead overall. Their top 10 by quantity is the same ten products as the overall list. By revenue, nine of ten matches. For every product in their top 10, repeat customers account for between 61% and 94% of its total net revenue. The lowest shares are party bunting (65%) and the Christmas paper chain kit (61%). This was not tested, but it may mean one-time customers buy more seasonal items.

Implication: keeping existing customers matters much more than winning new ones. Losing a few repeat buyers would cost more than losing many one-time buyers.

Recommendations:

Four recommendations follow from the findings above.

1. Protect returning customers first.
2. Prepare for autumn. Orders start climbing in September and peak in November, so stock and staff for more orders.
3. Grow abroad where it already works: the Netherlands, Ireland, Germany, France and Australia.
4. Judge products on net figures. Keep cheap high-volume items and a few higher-priced earners, and investigate why two huge orders were cancelled, because a pattern like that could point to a problem worth fixing.

Limitations and further questions:

- The data covers about one year, so a yearly pattern cannot be confirmed. The link to Christmas rests on timing alone.
- December 2011 has only 9 days of data, so its low total is not a fall in demand.
- About a quarter of the rows have no customer ID, so the customer question covers identified customers only.
- Many repeat buyers are probably wholesalers, and customers who first ordered late in the period may be counted as one-time.
- Revenue per order is approximate. Revenue is net of returns, while the order count leaves out cancellations, and a return counts in the month it was cancelled.
- The price analysis covers only the 50 best sellers, so it says nothing about slow sellers.

Analytical decisions and corrections:

Several checks changed the analysis along the way.

- Totals were checked before results were trusted. After each pivot, its grand total was compared with a direct sum of the column.
- Gross figures overstated two products, so the product analysis was restated net of returns.
- The first monthly line chart was ordered by revenue, which broke the time order and showed a steady decline that was not real. It was rebuilt in date order, and December 2011 is marked as a partial month.
- A hypothesis was tested and rejected. Bigger orders from wholesalers were expected to explain the autumn rise; the data showed more orders instead.
- Each answer was followed by the question of what a business would do with it, which is why every finding ends with an implication.